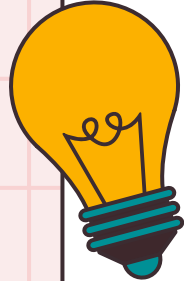
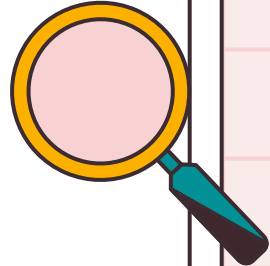


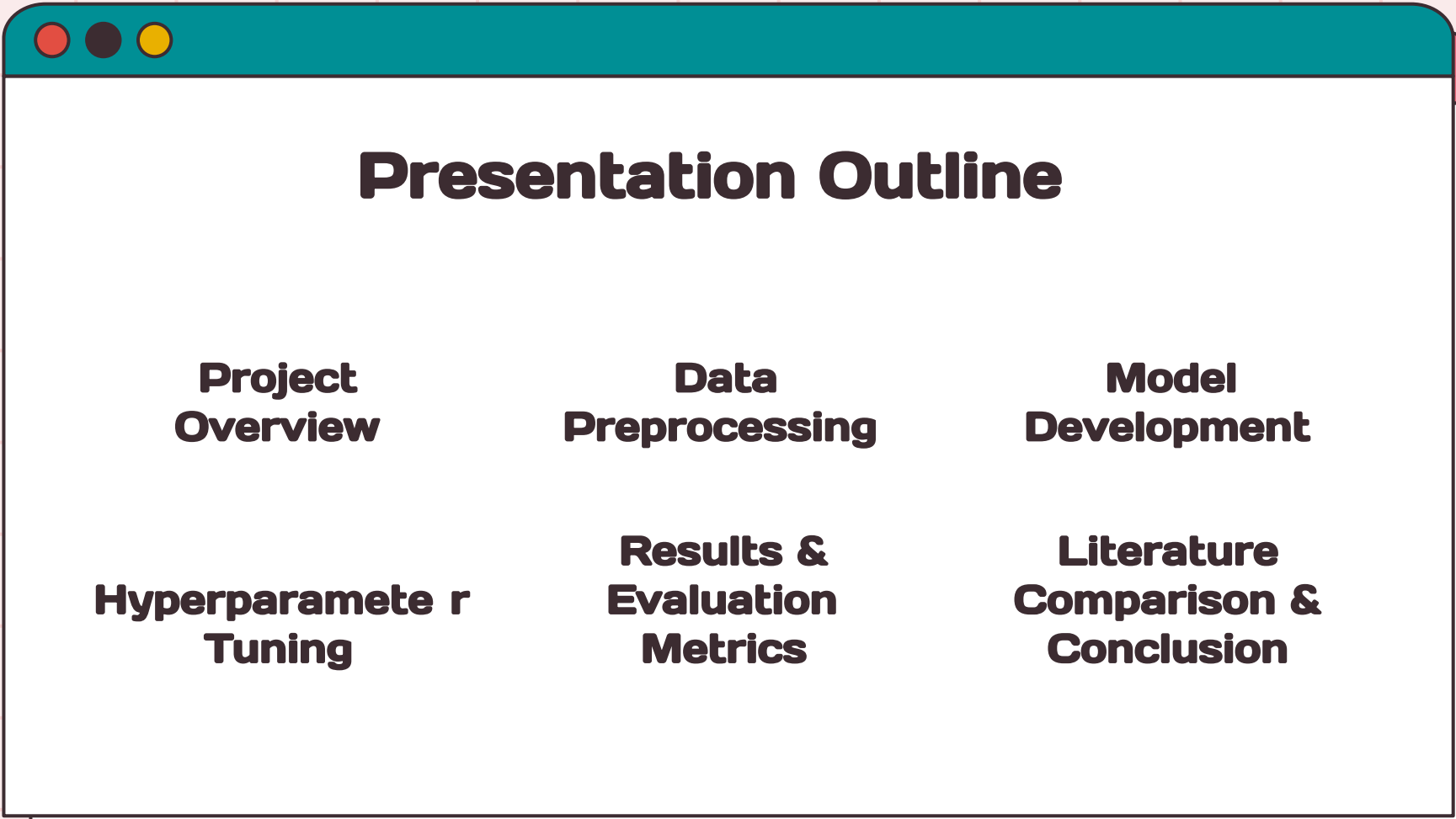


Final Project

Titanic & California Housing



Noah Wiley | CSCI 164: Artificial Intelligence Programming | Spring 2026

The slide is styled to look like a window with a teal header bar containing three colored circles (red, dark grey, yellow). The background of the slide is white with a light pink grid pattern. The title 'Presentation Outline' is centered at the top in a large, bold, dark grey font. Below the title, the content is organized into two rows of three items each, separated by vertical lines. The first row contains 'Project Overview', 'Data Preprocessing', and 'Model Development'. The second row contains 'Hyperparameter Tuning', 'Results & Evaluation Metrics', and 'Literature Comparison & Conclusion'. All text is in a bold, dark grey font.

Presentation Outline

**Project
Overview**

**Data
Preprocessing**

**Model
Development**

**Hyperparameter
Tuning**

**Results &
Evaluation
Metrics**

**Literature
Comparison &
Conclusion**

Datasets

Dataset 1: Titanic Survival

Binary Classification

- 891 passengers, 38.4 survival rate
- Target: survived (0 = died, 1 = survived)
- Feature: age, sex, pclass, fare, embarked
- Missing age/fare → median imputation
- Engineered features: family_size, is_alone
- 80/20 stratified train/test split

Dataset 2: California Housing

Regression Task

- 20,640 districts from 1990 U.S Census
- Target: MedHouseVal (in \$100,000)
- 8 numerical features, no missing values
- Key predictor, median income (Medlinc)
- StandardScale applied in pipeline
- 80/20 train/test split, random_state = 42)



Model Development

Logistic / Linear Regression:

Interpretable linear model. Best classifier on Titanic: 81.5 accuracy, AUC 0.865. Minimal assumptions, fast training, calibrated probabilities.

Decision Tree (Classifier & Regressor):

Recursive feature partitioning. Captures non-linear patterns. Prone to overfitting, hyperparameter tuning essential for both tasks.

k-Nearest Neighbors (Classifier & Regressor):

Non-parametric, instance-based learning. No training phase. Sensitive to feature scale, all models wrapped in StandardScaler Pipeline.



Hyperparameter Tuning - GridSearchCV

Decision Tree

Parameters searched:

- max_depth: [3, 5, 7, 10, None]
- min_samples_split: [2, 5, 10]
- min_samples_leaf: [1, 2, 4]

Best clf: depth=5, split=10, leaf=1
CV accuracy: 0.8158

Best reg: depth=10, split=2, leaf=4
CV R^2 : 0.7099

k-NN

Parameters searched:

- n_neighbors: [3, 5, 7, 9, 11, 15]
- weights: [uniform, distance]
- metric: [euclidean, manhattan]

Best clf: k=9, uniform, manhattan
CV accuracy: 0.8115

Best reg: k=9, distance, manhattan
CV: R^2 : 0.7313

Results - Titanic Classification

Model	Acc	F1	AUC	CV
Logistic Regression	0.815	0.744	0.865	0.792±0.016
Decision Tree	0.747	0.672	0.728	0.774±0.021
k-NN	0.781	0.702	0.832	0.808±0.020
Decision Tree (Tuned)	0.770	0.655	0.825	0.818±0.029
k-NN (Tuned)	0.787	0.721	0.846	0.799±0.031

California Housing Regression

Model	MAE	RMSE	R ²	CV
Linear Regression	0.533	0.746	0.576	0.553±0.062
Decision Tree	0.454	0.703	0.623	0.355±0.085
k-NN	0.446	0.658	0.670	0.521±0.044
Decision Tree (Tuned)	0.431	0.639	0.689	0.520±0.103
k-NN (Tuned)	0.406	0.603	0.722	0.562±0.47



Comparison To Prior Work

VanderPlas (2016) - Python Data Science Handbook

Their result: 81% accuracy — Logistic Regression on Titanic

My Result: 81.5% accuracy

Géron (2019) - Hand-On ML with scikit-learn

Their result: RMSE ~0.72(Linear), ~0.71(DT Tuned)

My Result: 0.746 / 0.639 RMSE

Kaggle Top Entries - Titanic Copetition Leaderboard

Their result: 83-86% using Random Forest / XGBoost

My Results: 81.5 % — gap expected, no ensembling

Conclusion

Best Classifier: Logistic Regression — 81.5%, AUC 0.865

Best Regressor: k-NN (Tuned) — R^2 0.722, RMSE 0.603

Biggest Improvement: Decision Tree Regressor CV R^2 0.355 $\rightarrow$ 0.520 after GridSearchCV

Best Practice: All models used Pipelines — no data leakage across CV folds

Future Work: Ensemble methods (Random Forest, XGBoost) would close the gap